

Saad Rahman Warsi

Software Engineer | Full-Stack | APIs | Backend | Cloud | DevOps

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Security Clearance Available: **Reliability**

PROFESSIONAL SUMMARY

- Result-driven Software Engineer (University of Ottawa | Jan'2026 | CGPA: 9.55/10) with a strong background building REST APIs, automating CI/CD pipelines, containerizing applications, and supporting scalable systems. Proficient in Java, Python, C, cloud platforms (GCP, Azure, AWS), Docker, Kubernetes, DevOps with hands on experience in embedded systems, real-time software design, and microcontroller-based development.
- Currently working in Agriculture and Agri-Food Canada since graduating in Jan'2026. In addition, gained experience through 4 Co-ops (16 months Full-Time) plus Part-Time experience (12 months) at Agriculture and Agri-Food Canada (AAFC), Canada Revenue Agency & CNW Group (Cision). Hold certifications in Azure and Scrum Master. Known for reliability, clean code, and the ability to contribute quickly to production systems.

TECHNICAL SKILLS

- Programming Languages: Java, Python, C/C++, SQL, Bash
- Backend Technologies: Spring Boot, REST APIs, Microservices, Node.js
- Frontend Technologies: Angular, React, Android (Java/Kotlin), HTML, CSS, JavaScript, TypeScript
- Cloud & DevOps: Docker, Kafka, SCDF, Kubernetes, GCP, Azure, AWS, OpenShift, Jenkins, CI/CD
- Databases: PostgreSQL, SQLite, Firebase, TileDB
- AI/ML Frameworks: PyTorch, Scikit-learn, Claude (prompt engineering, workflow automation)
- Embedded Systems: STM32, Arduino, Intel Quartus, RTOS Concepts, Scheduling, ISR Design
- Testing & Quality: Selenium, Pytest, JMH, PitTest, SonarQube
- Tools & Platform: Git, Linux, Maven, Jira, Confluence, VS Code, GitHub Copilot (AI-assisted coding)

PROFESSIONAL EXPERIENCE

Research Assistant (EG03, Contract, Full-Time) – Agriculture and Agri-Food Canada

May 2024 – Present | Science and Technology Branch, Ottawa, ON

- Developed and maintained backend services supporting research and analytics workflows
- Built genomic data storage solutions using SQLite, PostgreSQL and TileDB
- Deployed software tool PHG, which could potentially reduce per-sample processing costs from **\$500 → \$15**, scaling predictions from **300K → 34M** at **92%** accuracy using PostgreSQL, Docker, Singularity, and automated shell workflows.
- Achieved **98% & 92% imputation accuracy** on large and complex genomic datasets of 2 crops.
- Engineered fully automated shell scripts and Slurm pipeline with dependency chaining, reducing manual effort and enabling overnight unattended runs, reducing effort by **70%**
- Operated on HPC infrastructure managing **100 TB** of storage, optimizing workflows for large-scale genomic datasets.
- Developed containerized deployments ensuring reproducibility and portability across compute environments using Docker and Singularity

IT Programmer/Analyst (Co-op) – Canada Revenue Agency (CRA)

Sep 2023 – Dec 2023 | Ottawa, ON

- Developed DevHub, an internal collaborative platform, using Java, Spring Boot, Angular, Node.JS, and PostgreSQL, improving peer collaboration and issue resolution.
- Designed and developed OCBusTracker, a microservices-based web application with Angular frontend, containerized with Docker and deployed on OpenShift.
- Acted as Scrum Master for a sprint; enhanced Docker training curriculum and gained experience in Agile methodologies using JIRA.
- Collaborated with cross-functional teams to meet government standards.

Software Developer Intern – CNW Group (Cision)

Jan 2023 – Apr 2023 | Ottawa, ON

- Worked on Content Ingestion project responsible for data ingestion, enrichment, and delivery to output pipeline for analysis and operations.
- Implemented automated alerting for build failures, significantly cutting downtime and response time.
- Maintained cloud-based data streaming and supported migration to Google Cloud Platform (GCP).
- Resolved 4+ critical build issues, fixed a stream giving errors, tested a key component, and customized GCP.
- Strengthened application reliability through code reviews and debugging; worked in Agile teams using Git, Jira, and Confluence.

EDUCATION

- Bachelor of Applied Science (BASc) – Software Engineering (CO-OP)
- **University of Ottawa**, Ottawa, ON
- Graduated: January 2026 | **CGPA: 9.55 / 10**
- Dean's Honour List (2021–2025)
- University Merit Scholarship (2021–2025)

SELECTED PROJECTS

- Alarm System with Motion Sensor (STM32, C): Designed and implemented an embedded alarm system using an STM32 microcontroller and a motion sensor.
- Design and Simulation of a Basic Computer in Intel Quartus
- Ivy, an AI agent that autonomously completes medical intake forms. Built using Claude and Google's Gemini model, Ivy reads form structures at runtime, interprets labels and inputs, and drives a real browser to fill and submit them. Video link: <https://youtu.be/Rz6v5jclJjU>
- [Mathemagical Shapeplay](#) (React, JavaScript, Bootstrap): Developed an interactive web project that generates beautiful animations and patterns derived from mathematical equations.
- Establishing a DevOps Pipeline, video link: <https://youtu.be/Zm2-e2K-BQ8?si=biN6cdQZRqISCGfg>
- PharmFinder (Capstone, React, Spring Boot, PostgreSQL, Kubernetes): Enhanced the recruitment platform for Canada's leading pharmacy talent acquisition firm DPR Group Inc.
- Testing Project (Selenium, JMH, PitTest): Developed and executed automated test cases for an 'Online Book Store' web app, including performance, mutation, and coverage analysis. https://youtu.be/Mh8AdUPJ4io?si=5KrD0jvTxM_eHNIt
- Mealer (Android, Java, Firebase): Developed a meal-ordering app for local cooks to sell meals to clients from their homes.

CERTIFICATIONS

- [Microsoft Azure Fundamentals \(AZ-900\)](#), 2024
- [Professional Scrum Master I \(PSM-I\)](#), 2024

LANGUAGES

- English: Fluent
- French: Intermediate
- Hindi: Native